UNIT 5 · 9 hours

Summing Amplifier



Op-Amp and Linear IC Applications

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Draw an inverting summing amplifier
- Derive the output expression
- Design an averaging circuit
- Solve numericals

Syllabus context

Introduction, op-amp specifications, differential and common-mode operation, op-amp applications: inverting/non-inverting amplifier, summing, integrator, differentiator, comparator, 555 timer IC in astable mode, 78XX series IC voltage regulators.

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Circuit and operation

- Several inputs v1, v2, v3 connected through R1, R2, R3 to the inverting node.
- Non-inverting terminal grounded; feedback resistor Rf.
- The inverting node is a virtual ground, so each input current is independent: i1 = v1/R1, etc.
- KCL at the node: if = i1 + i2 + i3 and if = −vo/Rf.

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Output expression

$$v_o = -\left[\left(\frac{R_f}{R_1}\right) v_1 + \left(\frac{R_f}{R_2}\right) v_2 + \left(\frac{R_f}{R_3}\right) v_3 \right]$$

weighted sum

$$R_1 = R_2 = R_3 = R: \; v_o = -\left(\frac{R_f}{R}\right)(v_1 + v_2 + v_3)$$

scaled adder

$$R_f = R/3: \; v_o = -(v_1 + v_2 + v_3)/3$$

averager

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Numerical: adder

Problem R1 = R2 = R3 = 10 kΩ, Rf = 10 kΩ; v1 = 1 V, v2 = 2 V, v3 = 3 V. Find vo.

- $v_o = -(R_f/R)(v_1 + v_2 + v_3)$
- $= -(10/10)(1 + 2 + 3)$

Answer: $v_o = -6$ V

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Numerical: weighted sum

Problem R1 = 10 kΩ, R2 = 20 kΩ, R3 = 40 kΩ, Rf = 40 kΩ; inputs 1 V each. Find vo.

- Gains: Rf/R1 = 4, Rf/R2 = 2, Rf/R3 = 1
- $v_o = -(4 \times 1 + 2 \times 1 + 1 \times 1)$

Answer: vo = −7 V

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Applications

- Audio mixers: combine several channels.
- Averaging circuits (Rf = R/n).
- D/A converters using binary weighted resistors.
- Analog computers: summation block.

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Key takeaways

- $v_o = -(R_f/R_1 v_1 + R_f/R_2 v_2 + R_f/R_3 v_3)$
- Inputs do not interact because of the virtual ground
- Equal R and Rf = R/n gives an inverting average
- Basis of weighted-resistor DACs and mixers

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Practice questions

Q1. Derive the output of a 3-input summing amplifier.

Q2. Design an averager of three inputs.

Q3. R = 10 kΩ, Rf = 20 kΩ, v1 = 0.5 V, v2 = 1 V: find vo.

Q4. Why do the sources not interact in the summing amplifier?